Family Formation Readiness
Assignment 3 Integration Draft

Project team

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Integration status. This draft contains Fangzhou's revised contribution. Alanoud's descriptive section, Moza's predictive validation and the final combined discussion remain to be added. All supplied records are synthetic. The original generation procedure was not supplied.

1 Results

Alanoud's descriptive results are pending.

1.1 Factors associated with overall readiness

The primary model is a proportional-odds ordinal logistic regression of B10 using C1, C9, C14, E3, E13 and F9, with nationality, gender, role, age group, questionnaire language and engagement status as covariates. Of the 691 synthetic respondents with a reported nationality, 407 were single or engaged and formed the primary population; 11 who did not report their gender or age group were excluded, leaving 396 respondents (289 Emirati and 107 non-Emirati, of whom 34 were engaged). Because only 30 single or engaged respondents were aged over 30, and only seven of them were Emirati, the three oldest age groups were combined into a single 31-or-above category. Odds ratios (ORs) refer to the odds of a higher B10 category, and probabilities were obtained by averaging model predictions over respondents [1]. Because the dataset was generated by the research team to mirror the planned questionnaire, the estimates below show what the analysis reports when applied to data of this structure; they are not evidence about the Khalifa University population.

Table 1 reports the primary model. The six core items jointly improve the fit after nationality and the controls are included (likelihood-ratio $\chi^2(6) = 61.7$, $p < 0.001$), and the full model improves the log-likelihood by 13.9% relative to a threshold-only model (McFadden $R^2 = 0.139$). Financial preparedness (C14) has the strongest association. Each one-category increase in agreement with "I feel financially prepared to establish a family" multiplies the odds of a higher readiness category by 1.83 (95% CI 1.51–2.21) and, in a capped one-category prediction scenario, changes the average predicted probability of agreeing or strongly agreeing with B10 (categories 4–5) by 10.6 percentage points. Housing access (C9; OR 1.40, 1.16–1.69), family support (E3; OR 1.31, 1.07–1.60) and employment confidence (C1; OR 1.27, 1.02–1.57) form a second group, with coefficients 39–56% as large as that of C14 on the log-odds scale and probability changes of 3–5 percentage points.

E13 ("Greater confidence in my future would encourage me to establish a family") and F9 ("Better work-life balance would encourage earlier family formation") provide no clear evidence of association with readiness in this synthetic sample, with ORs of 1.04 (0.81–1.33) and 1.00 (0.78–1.28). These two items ask whether a hypothetical improvement would encourage family formation, whereas C1, C9, C14 and E3 describe the respondent's present circumstances. Agreement with F9 is high in both groups (mean 3.8 for Emirati and 4.1 for non-Emirati respondents on the five-point scale). The generated work-life balance responses illustrate a high stated preference with little estimated

conditional association. This does not justify a real-world policy priority or establish an absence of association.

Figure 1 presents these estimates together with the nationality term. The literature provides context for examining economic preparedness and family support [2–5]. Agreement between these synthetic patterns and earlier studies is not independent confirmation: the data-generation mechanism has not been supplied with the workbook, so it is not possible to establish which patterns were deliberately encoded.

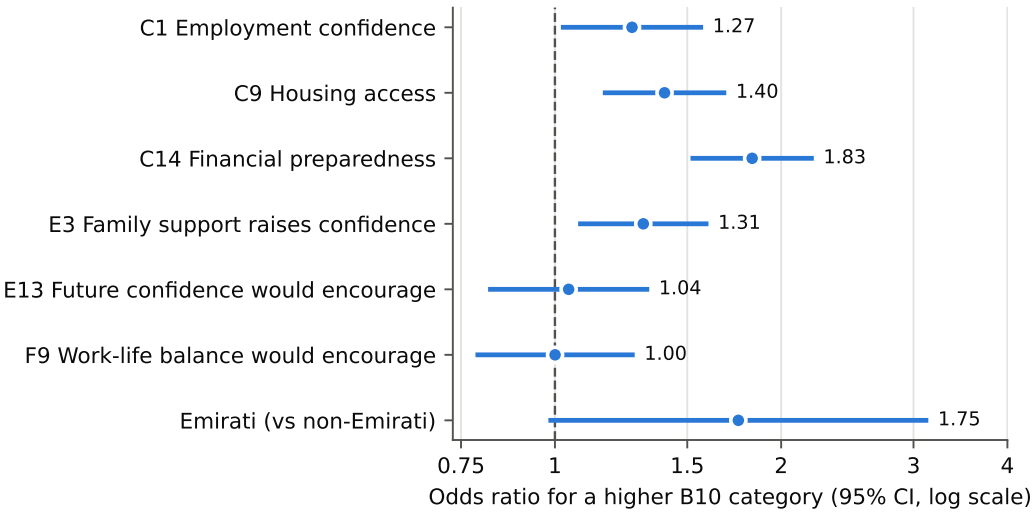


Figure 1: Odds ratios with 95% confidence intervals for the six core items and nationality, from the primary model in Table 1. Values to the right of the dashed line indicate higher odds of a higher B10 category.

Among the controls, engaged respondents have 4.7 times the odds of a higher readiness category than single respondents (2.3–9.6), and respondents aged 26–30 have 3.5 times the odds of those aged 18–20 (1.7–7.0). The engagement association is expected, since becoming engaged is itself partly an expression of readiness, and it should not be read as an effect of engagement. Gender (OR 0.77, 0.52–1.14), student or employee role (1.08, 0.50–2.36) and questionnaire language (Arabic versus English, 1.36, 0.87–2.11) show no clear association.

1.2 Comparison of Emirati and non-Emirati respondents

Emirati respondents report higher readiness than non-Emirati respondents before adjustment (mean B10 3.37 versus 2.98). After adjustment, the nationality OR is 1.75, but its interval includes one (0.98–3.14; $p = 0.058$). The groups also differ in housing access (mean 3.61 versus 2.64) and confidence from family support (3.75 versus 2.99). These summaries do not decompose or explain the nationality difference. Table 2 expresses the adjusted difference as predicted response distributions for the 378 respondents whose role and age-group combination is represented in both groups. With all other covariates held at their observed values, the predicted probability of agreeing or strongly agreeing with B10 is 38.9% when every respondent is treated as Emirati and 29.1% when every respondent is treated as non-Emirati. The difference of 9.9 percentage points has a 95% interval of 0.1 to 19.3, and Figure 2 shows that it consists mainly of a shift from category 2 to categories 4 and 5. The lower limit of this interval lies just above zero while the OR interval just includes one; the boundary-sensitive result should not be treated as a firm finding. The interval uses only 1,000 parameter draws and is conditional on the selected overlap set; Monte Carlo variability and overlap selection are not included. The role-by-age restriction does not establish overlap in language or all six predictors. With 107 non-Emirati respondents, the data cannot estimate the adjusted nationality

Table 1: Primary ordinal logistic regression of overall readiness (B10) among single and engaged respondents ($n = 396$).

Term	$\hat{\beta}$	SE	OR	95% CI (OR)	p
Core items (1–5 agreement, per one-category increase)					
C1 Employment confidence	0.236	0.111	1.27	[1.02, 1.57]	0.034
C9 Housing access	0.336	0.096	1.40	[1.16, 1.69]	<0.001
C14 Financial preparedness	0.604	0.096	1.83	[1.51, 2.21]	<0.001
E3 Family support raises confidence	0.271	0.102	1.31	[1.07, 1.60]	0.008
E13 Future confidence would encourage	0.042	0.126	1.04	[0.81, 1.33]	0.738
F9 Work-life balance would encourage	0.000	0.124	1.00	[0.78, 1.28]	0.997
Emirati (vs non-Emirati)	0.562	0.297	1.75	[0.98, 3.14]	0.058
Controls (references: female, student, 18–20 years, English, single)					
Gender: male	-0.265	0.202	0.77	[0.52, 1.14]	0.190
Role: employee	0.079	0.399	1.08	[0.50, 2.36]	0.843
Age: 21–25	0.254	0.226	1.29	[0.83, 2.01]	0.262
Age: 26–30	1.246	0.361	3.48	[1.71, 7.05]	<0.001
Age: 31 or above	0.982	0.560	2.67	[0.89, 8.00]	0.080
Language: Arabic	0.304	0.225	1.36	[0.87, 2.11]	0.177
Engaged (vs single)	1.554	0.363	4.73	[2.32, 9.63]	<0.001
Log-likelihood -428.4 (threshold-only -497.4); LR $\chi^2(14) = 138.1$, $p < 0.001$; McFadden $R^2 = 0.139$					
Six core items jointly: LR $\chi^2(6) = 61.7$, $p < 0.001$; controls jointly: $\chi^2(7) = 49.9$, $p < 0.001$					

Note: OR > 1 indicates higher odds of a higher B10 category. Thresholds: $\tau_1 = 0.75$, $\tau_2 = 3.83$, $\tau_3 = 6.43$, $\tau_4 = 9.00$. Respondents reporting both roles are coded as students. Estimates are from synthetic data and are not population estimates.

difference precisely.

Table 2: Covariate-standardised probability (%) of each B10 response for Emirati and non-Emirati respondents on the common-support set ($n = 378$).

B10	Emirati	Non-Emirati	Difference (pp)
1	1.0 [0.4, 2.5]	1.8 [0.7, 4.7]	-0.7 [-2.5, -0.0]
2	14.8 [11.8, 19.7]	21.5 [16.3, 31.2]	-6.7 [-15.3, -0.1]
3	45.3 [40.6, 49.7]	47.6 [42.0, 52.0]	-2.4 [-4.0, +0.3]
4	31.5 [26.1, 35.7]	24.4 [17.9, 30.9]	+7.1 [+0.1, +14.3]
5	7.4 [5.0, 10.1]	4.7 [2.6, 7.5]	+2.7 [+0.0, +5.4]

Note: Each respondent's covariates are retained and nationality is set to each value in turn. The common-support set contains the role × age-group cells with at least three respondents from each group (employee, 26–30; employee, 31 or above; student, 18–20; student, 21–25; student, 26–30). 95% intervals in brackets from 1000 draws of the estimated parameter distribution. B10: "Overall, I consider myself ready for family formation" (1 = strongly disagree, 5 = strongly agree).

The secondary model allows each core item to have a different association in each group. The six nationality × item interactions jointly improve the fit ($\chi^2(6) = 18.6$, $p = 0.005$), and Table 3 shows where the differences lie. The clearest difference concerns family support. Among Emirati respondents, each one-category increase in E3 multiplies the odds of higher readiness by 1.66 (1.30–2.14), whereas among non-Emirati respondents the OR is 0.84 (0.60–1.19; difference $p = 0.002$). Financial preparedness shows the opposite pattern, with an OR of 2.48 (1.75–3.53) for non-Emirati and 1.63 (1.31–2.04) for Emirati respondents ($p = 0.043$). Housing access is associated with readiness among Emirati respondents (1.64, 1.30–2.07) but less clearly among non-Emirati respondents (1.13, 0.79–1.63; $p = 0.089$). These item-level comparisons are exploratory and are not adjusted for multiple testing.

Family location suggests a hypothesis for future empirical investigation, rather than an explanation established by these synthetic records. Of the Emirati respondents in the primary population, 95.5%

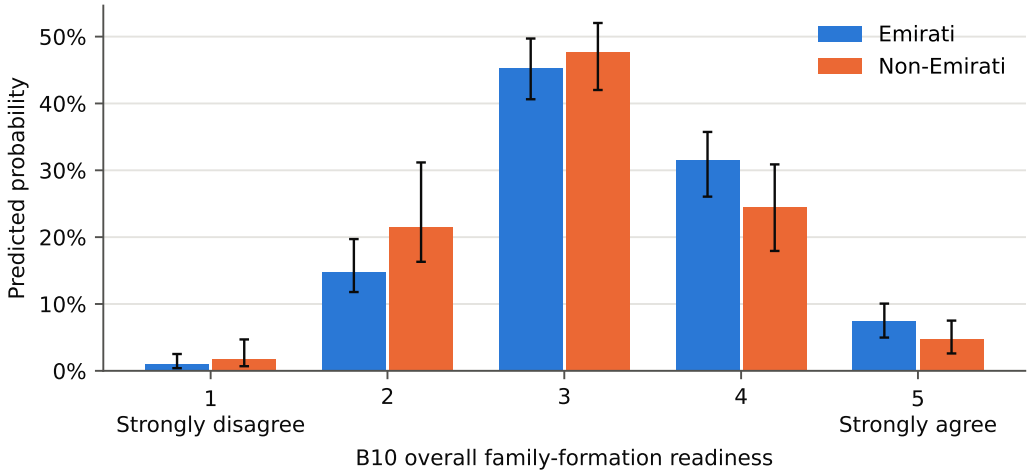


Figure 2: Covariate-standardised probability of each B10 response for Emirati and non-Emirati respondents with 95% intervals, on the common-support set (values in Table 2).

report that their family lives in the UAE, compared with 21.5% of non-Emirati respondents; 56.1% of non-Emirati respondents report a family living entirely outside the country. Family support can be translated more readily into practical help with housing, childcare and wedding arrangements when the family is nearby, while respondents without such support may depend more on their own savings. This would also be consistent with the larger financial-preparedness association among non-Emirati respondents. Family location was not part of the prespecified model, and the explanation therefore remains a hypothesis for the collected data to test.

Table 3: Odds ratios of the six core items for Emirati and non-Emirati respondents, from the secondary model with nationality × item interactions ($n = 396$).

Core item	Emirati OR [95% CI]	Non-Emirati OR [95% CI]	p (difference)
C1 Employment confidence	1.18 [0.91, 1.53]	1.39 [0.92, 2.09]	0.510
C9 Housing access	1.64 [1.30, 2.07]	1.13 [0.79, 1.63]	0.089
C14 Financial preparedness	1.63 [1.31, 2.04]	2.48 [1.75, 3.53]	0.043
E3 Family support raises confidence	1.66 [1.30, 2.14]	0.84 [0.60, 1.19]	0.002
E13 Future confidence would encourage	1.03 [0.77, 1.38]	1.01 [0.61, 1.66]	0.943
F9 Work-life balance would encourage	0.94 [0.71, 1.24]	0.86 [0.51, 1.46]	0.779

Note: Joint test of the six interactions: LR $\chi^2(6) = 18.6$, $p = 0.005$. Item-level p-values are exploratory and unadjusted for multiplicity. Controls as in Table 1.

1.3 Scenario predictions

Table 4 reports the prespecified Monte Carlo scenarios, which raise C1, C9 or C14 by one category for every respondent, with values capped at 5. Raising financial preparedness gives the largest change in the predicted share agreeing or strongly agreeing with B10, an increase of 11.2 percentage points for Emirati respondents (7.7–14.0) from a baseline of 41.9%, and of 8.7 points for non-Emirati respondents (5.8–11.4) from a baseline of 24.8%. Raising housing access adds about 5 points in both groups, and raising employment confidence adds about 3 points. The C1 and C9 scenarios for Emirati respondents are limited by a ceiling, because 27% and 22% of them already select the top category. Only 3% of non-Emirati respondents select the top housing category, which leaves more room for change in that group. These scenarios use the primary model with shared predictor slopes, not the interaction model. Differences between group-specific scenario changes can therefore arise from covariate distributions and ceiling effects. The scenarios use each group's own analysed records,

whereas Table 2 uses the pooled overlap set, so the baselines differ. No multidimensional support or substantive plausibility check of the shifted profiles has been established. They do not estimate the effect of providing housing, employment or financial support, because readiness may also influence how respondents assess their own circumstances.

Table 4: Monte Carlo scenarios: change in the mean predicted probability of agreeing or strongly agreeing with B10 ($\Pr(Y \geq 4)$) when one item is raised by one category for every respondent.

Scenario	Change in $\Pr(Y \geq 4)$, pp [95% interval]	Already at 5 (%)
Emirati ($n = 289$), baseline 41.9%		
C1 Employment confidence	+3.4 [+0.1, +6.3]	27
C9 Housing access	+5.2 [+2.3, +7.7]	22
C14 Financial preparedness	+11.2 [+7.7, +14.0]	9
Non-Emirati ($n = 107$), baseline 24.8%		
C1 Employment confidence	+2.7 [+0.1, +5.1]	21
C9 Housing access	+5.0 [+2.1, +7.6]	3
C14 Financial preparedness	+8.7 [+5.8, +11.4]	7

Note: Values are capped at 5, so respondents already at the top category are unchanged. Intervals reflect parameter uncertainty only (1000 draws). Scenarios compare predictions at different reported conditions; they are not effects of providing employment, housing or financial support.

1.4 Model diagnostics and robustness

Table 5 reports the diagnostics and sensitivity analyses. The archived custom Brant test is not used to establish proportional odds. Only five records, all non-Emirati, selected B10 = 1; the resulting cumulative binary fit is vulnerable to separation. Threshold-specific coefficient inspection and a defensible assumption assessment remain required. Variance inflation factors do not exceed 1.23 for the core items or 2.23 for any term, although low VIF does not establish conceptual distinctness. The archived category-coded sensitivity merges predictor categories 1 and 2, whereas the primary model retains their distinct numeric values. Its likelihood-ratio p-value has therefore been removed because the comparison is not nested. Its AIC (905.7 versus 892.8) is descriptive and does not validate linear item effects. In-sample modal accuracy is 51.5%, versus 45.2% for the majority-category rule, and 97.0% of fitted predictions are within one category. These are training-sample summaries; predictive validation and calibration on held-out records are reserved for Moza's contribution.

The core-item estimates are stable across specifications. In the wider sample of 655 respondents, which adds married and divorced or widowed respondents and adjusts for marital status, the housing association becomes larger (OR 1.65), while the other items and the nationality term change little. Merging the sparse B10 categories 1 and 2 changes no item OR by more than 0.02 and lowers the nationality OR from 1.75 to 1.57. The two support extensions provide limited evidence in their selected subsets. The D8.1 extension is limited to the 130 respondents (108 Emirati) who named a programme, reported being eligible and rated its usefulness. In this subset, the programme-usefulness rating has an OR of 1.16 (0.78–1.72) and does not improve the AIC of the core model on the same respondents (315.1 versus 313.7). Only 12 of 392 assessable respondents recognised none of the listed initiatives, and all 12 were non-Emirati. The awareness contrast consequently lacks within-Emirati support and is imprecise; awareness and nationality are not perfectly collinear because non-Emirati records include both recognisers and non-recognisers.

Table 5: Variance inflation factors and sensitivity odds ratios in synthetic data.

Term	VIF	Wider sample ($n = 655$)	B10 1–2 merged	Core, D8.1 subset ($n = 130$)
C1 Employment confidence	1.14	1.19	1.25	1.07
C9 Housing access	1.22	1.65	1.39	1.27
C14 Financial preparedness	1.11	1.70	1.81	1.46
E3 Family support raises confidence	1.15	1.25	1.32	1.33
E13 Future confidence would encourage	1.15	1.10	1.04	1.18
F9 Work-life balance would encourage	1.07	1.00	0.99	0.93
Emirati (vs non-Emirati)	1.83	1.76	1.57	1.05

The archived Brant p-values and non-nested likelihood-ratio test have been removed. Proportional odds remains unverified. Wider-sample and D8.1-subset estimates are retained from Fangzhou's supplied outputs.

Moza's out-of-sample validation is pending.

2 Discussion

Combine the interpretation with Alanoud's descriptive findings and Moza's validation before submission. The synthetic results do not establish population differences, programme effectiveness or bilingual measurement comparability.

3 Limitations

The ordinal results are subject to five limitations. First, the responses are synthetic and were supplied as a synthetic workbook; the generation script and assumptions were not included. Any association in Tables 1–5 therefore reflects, at least in part, the assumptions used to generate the data, and agreement with earlier UAE studies cannot be counted as independent confirmation. The analysis demonstrates what the prespecified model reports and where it lacks information; it does not estimate readiness in the Khalifa University community.

Second, the comparison between nationality groups is constrained by sample composition. The primary population contains 107 non-Emirati respondents, only three of whom selected the highest B10 category and five the lowest, and only seven Emirati respondents were aged over 30. The adjusted nationality difference is accordingly imprecise. Age groups above 30 had to be combined, and predictions were restricted to role and age combinations represented in both groups. In addition, 116 of 123 Arabic-language responses came from Emirati respondents, so the language and nationality terms are only weakly separable.

Third, the support-programme extensions could not answer their intended questions. Only 12 respondents, all non-Emirati, recognised none of the listed initiatives, so the awareness contrast lacks within-Emirati support and is imprecise. The usefulness extension relied on 130 respondents, 108 of them Emirati, who reported their own eligibility, and that eligibility was not independently verified.

Fourth, the design is cross-sectional and the outcome is a single self-reported item. Reported financial preparedness, housing confidence and readiness may share a common response tendency, and readiness may shape how respondents evaluate their own circumstances. The scenario results in Table 4 are therefore comparisons between predictions, not effects of providing support. Uncertainty intervals reflect parameter estimation only and do not include measurement error in B10 or the choice of specification.

Fifth, the interaction analysis tests six item-level differences. The joint test was prespecified, but the individual comparisons, including the stronger association of financial preparedness with readiness among non-Emirati respondents ($p = 0.043$), are exploratory and unadjusted for multiplicity. The family-location explanation for the difference in family-support associations was proposed after the results were seen and requires confirmation in the collected data.

The custom proportional-odds diagnostic has been withheld pending reliable threshold-specific assessment. In-sample accuracy is not external predictive validation. The overlap restriction uses role and age only, and the scenario profiles have not been validated for joint support or substantive plausibility. These limitations constrain interpretation even within the synthetic demonstration.

References

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